



Optimising the Late-Stage Rehabilitation and Return-to-Sport Training and Testing Process After ACL Reconstruction

Matthew Buckthorpe^{1,2}

Published online: 19 April 2019
© Springer Nature Switzerland AG 2019

Abstract

Despite increased knowledge on anterior cruciate ligament (ACL) injury mechanisms, improved surgical techniques, improved understanding of ACL biomechanics and enhanced knowledge in rehabilitation practice, return-to-sport (RTS) rates and subsequent second ACL re-injury rates after ACL reconstruction are not optimal. This narrative review discusses factors that may be highly relevant for RTS training and testing after ACL reconstruction, but which have received limited research attention to date or do not form part of the standard approach to rehabilitation. These factors include (1) explosive neuromuscular performance; (2) movement quality deficits associated with re-injury risk, particularly the need to re-train optimal sport-specific movement patterns; (3) the influence of fatigue; and (4) a lack of sport-specific re-training prior to RTS, with particular attention to an insufficient development of chronic training load. In addition, incorporating performance re-training and ensuring an athlete has restored their sport-specific profile is important. The relevance of these variables for RTS training and testing is discussed, with a new recommended model of late-stage rehabilitation and RTS training presented. Additional testing to support RTS decision making is also presented. This paper contains important information for practitioners and researchers to support optimised late-stage rehabilitation and RTS programmes and RTS testing with a view to enhancing patient outcomes after ACL reconstruction.

1 Introduction

The return to sport (RTS) outcomes following anterior cruciate ligament reconstruction (ACLR) are unsatisfactory. Most athletes who rupture their ACL are advised to undergo ACLR to continue to be able to take part in sport [1]. However, 35–45% of patients undergoing ACLR do not RTS [2, 3]. Even in elite sports environments, typically 20–25% of athletes are unable to RTS after ACLR [4, 5]. More importantly, those who RTS do so at unacceptable re-injury risk. The overall secondary ACL injury risk after ACLR is around 15% [6], whilst young athletes overall appear to have a 30% chance of re-injury within the first 2 years after RTS [7–9]. In relation to matched controls, this indicates a 30- to 40-fold heightened risk of re-injury upon RTS [6], which

is clearly unacceptable. As such, it appears that typically adopted functional recovery programmes and RTS criteria after ACLR need improvement. Returning an athlete back to sport after ACLR is complex, easy to do badly, whilst difficult to do well. Successful RTS requires a bio-psycho-social approach to rehabilitation and RTS [10] and is likely influenced by multiple factors. There is no consensus on rehabilitation and RTS models and criteria after ACLR [11, 12]. Current approaches fail to adequately identify and resolve residual deficits associated with explosive neuromuscular performance, movement quality, fatigue and sport readiness [13]. It is likely that most rehabilitation approaches are not comprehensive enough, do not provide sufficient intensity or are not specific enough to fully prepare an athlete for the demands of their sport. The primary purpose of this paper is to discuss important factors related to the late-stage rehabilitation and RTS training process and RTS testing process that have either received limited attention within the research or are not typically implemented within the functional recovery process after ACLR. These may be clinically important to optimise the RTS process and testing and for the prevention of re-injury after ACLR.

✉ Matthew Buckthorpe
M.Buckthorpe@isokinetic.com

¹ Isokinetic Medical Group, FIFA Medical Centre of Excellence, Education and Research Department, Bologna, Italy

² Isokinetic Medical Group, FIFA Medical Centre of Excellence, 11 Harley Street, London W1G 9PF, UK

Key Points

Despite advances in surgical techniques, return-to-sport and re-injury rates after anterior cruciate ligament reconstruction are disappointing, highlighting a need to optimise the functional recovery process.

Often patients return-to-sport with insufficient and sub-optimal training, reporting lower levels of performance, ongoing deficits in neuromuscular function and movement quality, and insufficient sport-specific conditioning.

There is a need to optimise the return-to-sport training process, ensuring sufficient training volume prior to return to sport to limit exposure to high acute:chronic workload ratios, thus reducing the possibility of elevated re-injury risk.

These important variables (neuromuscular function, movement quality, sport-specific conditioning, training load) need to be effectively implemented into a return-to-sport training framework (five stages), which should have a specific structure and involve on-field rehabilitation, gym-based re-conditioning and novel movement re-training services.

Return-to-sport testing, particularly in elite athletes, needs to be more comprehensive, incorporating more novel aspects such as assessment of neuromuscular function, movement quality, cardiovascular conditioning and assessment of training workloads.

2 Deficits in Explosive Neuromuscular Performance

The ability of the neuromuscular system to develop force is important to provide dynamic stability to a joint, as well as for optimal force propulsion. Typically, there is an over-reliance on isolated maximal muscle strength after ACLR, with limited consideration of the ability to develop force explosively [14]. Rapid (< 50 ms) re-stabilisation of joints following mechanical perturbation to prevent injury [15], or explosive athletic tasks such as sprint running (100–120 ms) [16], involve contraction times that are shorter than the time it takes to produce the maximal isometric voluntary force, which is typically in the order of 300 ms [17]. Hence, the capability to produce force during rapid sporting tasks (i.e. explosive efforts) may be more dependent upon the ability to increase force quickly from low levels, termed rate of force development (RFD), than on maximal muscle strength. As such, RFD appears to be an important aspect of neuromuscular function and may require additional consideration in late-stage rehabilitation programmes [14]. Angelozzi et al.

[18] reported deficits of RFD of 30% at 6 months following ACLR despite full restoration of maximal concentric knee extensor strength (97%). RFD was only restored after 12 months following a programme of power development, and the authors suggested RFD may be a useful adjunct measure to determine readiness to RTS after ACL injury.

RFD is influenced by different determinants depending upon the proportion of the force–time curve measured (i.e. 0–50 ms, 100–150 ms) [19, 20]. RFD after 150 ms from force onset is largely influenced by maximal muscle strength ($r^2=0.80$) [19, 21] and strongly associated with changes in maximal strength with training [20, 22, 23]. Early and mid-phase RFD (0–50 ms, 0–100 ms) are more closely associated with factors that influence relative RFD (the ability to produce a specific percentage of maximal force at specific time points [19]), including the level of agonist muscle activation at activation onset, the contractile properties of the muscle, muscle fibre type and the architectural characteristics of the muscle [20]. For example, maximal strength explained only 20% of the variability in force at 50 ms [19, 21], whilst agonist activation explained around 50% of the variability [19, 24] in recreational-level athletes.

It is recommended after achieving sufficient basic neuromuscular function after ACLR (e.g. minimum of at least 80–90% limb symmetry index [LSI] for isokinetic knee flexor and extensor strength [25–27]) that a period of additional advanced neuromuscular training be undertaken to restore explosive neuromuscular performance. Moderate-load resistance training (e.g. 70% maximal load that can be lifted for 8–12 repetitions), which is highly effective at developing muscle size and maximal strength [22, 23, 28], appears largely ineffective at training or restoring RFD [22, 29, 30]. Instead, specific stimuli are required to develop RFD over the initial 50–100 ms, including explosive strength training, which involves the rapid development of force from low to high levels, typically under isometric conditions [24, 31]; ballistic strength training [32], such as jumping or bounding [33]; and heavy (< 5 repetition maximum) strength training [30, 32]. Gains in RFD over the initial 50 ms of 50–80% can be expected after as little as 4–8 weeks of training with these modalities, at least in uninjured individuals [31, 32, 34]. Thus, late-stage rehabilitation should incorporate these specific training practices (explosive isometrics, ballistics, plyometrics) as part of a mixed-methods approach to restoring neuromuscular performance. It is important these are included alongside specific strength training involving an eccentric component [34] of moderate to high loads (e.g. 5–8 repetition maximum training) to fully restore maximal muscle strength. This should aim to fully restore (e.g. achieve a 100% LSI [25, 26]) open kinetic chain maximal strength of the knee extensors and flexors, as well as closed kinetic maximal strength (e.g. leg press strength of two times body mass, for 8 repetition maximum and 100%

LSI [35]). Typically, knee extensor maximal muscle strength is not restored at 6–9 months following ACLR in a large number of patients [36, 37]. As such, increased focus on neuromuscular performance as a whole (strength, power, RFD) appears warranted in order to optimise patient outcomes after ACLR. Obviously, high load/explosive strength and power training after injury must be proposed only after completion of the healing process, following recovery of range of motion and flexibility, and applying the principle of progressive overload [38].

In addition, specific assessment of RFD should be included in the ACL functional recovery pathway to support optimal criterion-based progression and RTS decision making. Testing RFD is experimentally challenging [14, 20], but by controlling certain factors that influence RFD performance such as countermovement, system compliance (the amount of compliance within the testing equipment) and athlete instructions [14, 20], it is possible to obtain valid and reliable information [39]. Assessment of closed kinetic chain RFD (e.g. leg press [18] or squat [40, 41]) explosive isometric contractions can provide useful information on the explosive strength characteristics of the lower limb and should be included where possible. However, it must be noted that multiple joint measures of RFD may not capture deficits specific to individual muscle groups (e.g. knee extensors or flexors). Thus, in addition to closed-chain RFD assessment, it is also advised to assess RFD during single-joint open kinetic chain tasks (e.g. explosive isometric knee extensors and flexor contractions) [42]. Force/torque should be assessed over the early (e.g. 100 ms) and late (200 ms) portions of the force/torque–time curve following force onset [19, 22, 34], alongside peak RFD and peak force. As there are no published recommendations for assessing RFD after ACLR, it is advised here to have a minimum of 90%, in line with research on maximal muscle strength testing prior to RTS training [25]. Maximising reliability by capturing a sufficient number of explosive contractions (e.g. eight explosive contractions with an emphasis on fast and hard [20, 23]) and taking the average of a number of trials (e.g. $n=3$) is recommended [39]. Further research is warranted to understand which specific cut-off criteria may be necessary for supporting RTS decision making.

When it is not possible or plausible to utilise a force plate or isometric strength testing rig and associated sophisticated software, it is advised to assess jump height and contact time during a single leg drop jump, measuring the reactive strength index (flight time/contact time) [35]. Ensuring a maximal jump is performed (to maximise power output) is important as there can be a tendency to minimise ground contact time at the expense of jump height, which can affect the validity of captured information [43]. Finally, as a minimum, assessment of countermovement performance, using

apps such as MyJump (which has recently been validated [44]) should be undertaken.

3 Deficits in ‘Movement Quality’

Residual movement impairments following ACLR are apparent during an array of functional tasks after ACLR [45–50]. ACL injury and subsequent reconstruction can lead to altered movement quality (defined here as the ability to control the limbs and achieve sufficient balance and kinematic alignment during functional activities, not displaying movement asymmetries or risk factors linked to ACL injury) of patients bilaterally [49], which has been prospectively linked with elevated secondary ACL injury risk [51, 52]. As such, movement re-training prior to RTS should form an important element of the functional recovery process. The need to restore movement quality after ACLR is nothing new [12, 53]. However, it must be noted that current practices in ACL rehabilitation typically fail to adequately optimise movement quality, given the residual movement impairments identified at the time of RTS [45–52]. The inability to correct movement issues prior to RTS is likely due to the use of incomplete programmes or a lack of volume and intensity of corrective movement re-training programmes. Movement is complex and influenced by an array of neuromuscular, biomechanical, sensorimotor and neurocognitive factors [12, 54, 55]. Based on the literature and in the author’s experience, there is a need to incorporate a holistic systematic movement re-training programme to address all the factors that may impact movement quality and ensure transfer (to sport-specific scenarios) and retention of movement patterns [56].

Firstly, neuromuscular and biomechanical factors such as muscle strength imbalances/agonist weakness (e.g. knee extensor weakness [57] and altered reciprocal muscle inhibition [58]) and neuromuscular activation (e.g. altered timing of muscles, such as delayed or reduced activation of medial hamstrings [59]) may affect movement quality. Assessing and resolving these factors are important initial steps in the movement retraining process and a well-designed neuromuscular training programme can support enhanced movement quality [60] and ACL injury risk reduction [61]. However, most of these factors should be addressed within mid-stage rehabilitation.

Secondly, alterations in neuromuscular function do not transfer directly to enhanced expression of movement quality while performing functional type tasks. Instead, there is a need to incorporate movement practice to relearn and optimise movement coordination during these tasks [62]. The training should include movement coaching to support cognitive understanding of the movement tasks [63], using biofeedback techniques such as mirror training, video

feedback and dyad training [64], and encourage an external as opposed to internal focus of attention [65]. It is also important to incorporate optimal movement progressions of progressive increase in loading through systematic increases in movement complexity [66] and speed [67], ensuring optimal 'coach' feedback (correct timing, content, amount of information). These movements should be relearned under both controlled tasks in the gym (or specialist movement environment) and during linear and then multi-directional movements on the field. In addition, confirming movement quality using two-dimensional (2D) video analysis and qualitative movement assessment in an array of 'sport-type' tasks is recommended (e.g. squat, landing, jumping, decelerating, changing direction) [12].

Finally, and most importantly in the context of this paper, there are various additional sport-specific factors commonly included in or assessed as part of a movement re-training approach after ACLR. Although optimising patients' movement quality during movements such as squatting, jumping, landing and change of direction tasks, and confirming this with 2D visual movement assessments [12], is an essential improvement in the traditional ACL rehabilitation approach, it is important to note that ACL injuries in sport do not typically occur during these 'pre-planned' movement tasks [68]. Rather, ACL injuries occur during the performance of highly complex tasks [15, 68]. Specifically, ACL injuries are generally thought to occur as a result of failure to maintain neuromuscular control whilst attending an external focus of attention, involving highly complex dynamic visual stimuli, variable surfaces, movement planning and rapid decision making with variable player positioning and environmental interaction during unanticipated movements [68]. There are many contextual differences between the 'pre-planned' lab type movements and the actual movement demands during sporting activity, which may limit the success of traditional movement re-training and testing approaches and include the following:

1. ACL injuries are now becoming recognised as neurophysiological injuries [12, 69] and alterations in neurocognitive and sensorimotor control after ACL injury, reconstruction and rehabilitation are important [12, 55, 70]. During sporting participation, the neurocognitive system of the athlete (e.g. situational awareness, sensory integration, motor planning and coordination) behaves collectively and interacts with its environment [55]. An athlete has to visually perceive the highly complex unpredictable environment (e.g. movement of another player, opponent, or a ball), which is constantly changing, quickly process these situation-specific visual-spatial cues within the central nervous system, and develop an appropriate movement strategy for the specific situation. Failure to search and process the environmental

information may result in deficits in performance and/or injury [70]. Disruption to the native ACL after injury leads to mechanical instability of the knee and can alter neuromuscular control due to disrupted mechanoreceptors within the ligament [71] and altered somatosensory input and joint proprioception. This decreased somatosensory input requires the patient to increase their reliance on visual feedback and increased conscious cortical involvement to effectively regulate optimal motor control [69]. Although this conscious overriding appears to be effective during simple or pre-planned motor tasks typical of most movement tests, it may not be effective during movements typical of sporting-specific tasks (i.e. reactive change of direction tasks or opposed drills such as defensive play in football), as the brain may become overwhelmed and less efficient, resulting in increased injury risk [69]. It appears that the addition of experimental, visually cued temporal constraints to movement testing can affect movement quality and knee loading during sport-type movements, such as cutting [72, 73], and that the addition of 'cognitive load' during postural control tasks negatively affects performance, with significantly more marked deficits in ACLR patients than healthy controls [74–76]. In addition, reactive movements induce worse biomechanics and higher knee loads than planned movements, with knee loads being twice as high during unplanned than planned side-stepping in the frontal and coronal plane [66]. Thus, re-training reactive movement quality and confirming quality of movement during reactive movement tasks prior to RTS appears an important and often neglected element of RTS training.

2. Movement testing prior to RTS is often performed when the patient is non-fatigued. Although not a consistent finding [77, 78], numerous laboratory-based studies have reported that acute fatigue results in reduced neuromuscular function and altered biomechanics [79–83]. The studies that reported an association between fatigue and altered movement quality showed fatigue to manifest in numerous ACL biomechanical risk factors during functional tasks, such as increased peak proximal tibial anterior shear force and decreased knee flexion angles [82], decreased hip flexion angle [83], increased knee valgus [83, 84], and increased knee internal rotation [85] during various athletic sporting movements. Furthermore, the combination of fatigue and unanticipated movement (typical of team sports) further results in heightened 'at risk' ACL biomechanics [82]. Thus, practicing movements under fatigue and confirming movement quality whilst fatigued are important additional elements of the movement re-training programme.

In summary, restoring movement quality is a vitally important element of ACL rehabilitation. Addressing the

biomechanical, neuromuscular, sensorimotor and neurocognitive elements of movement are important. Recognising the required movement patterns of the athlete's sport and ensuring a staged and multi-targeted approach to movement re-training is recommended to optimise athletes' movement quality upon RTS. In the author's experience, this should include and progress by (1) addressing the neuromuscular and biomechanical factors that affect movement quality and motor learning, (2) including a progressive movement re-training approach to re-learn an array of functional tasks optimising coordination and motor learning and (3) performing the final aspect of rehabilitation and movement training on the field, in realistic environments, progressively simulating the sporting movement demands and environmental constraints. This can be achieved through a sport-specific movement programme, with gradual increases in sport specificity through the provision of a greater number of options (e.g. three passing options as opposed to one) and movement choices, opponents and distractions to prepare the athlete for the chaotic and highly complex demands of their sport. Although screening should occur throughout the functional recovery approach according to criterion-based rehabilitation, RTS testing should consider incorporating qualitative movement assessment of functional tasks during both planned and reactive scenarios (e.g. cutting), whilst not fatigued and fatigued, as well as assessing movement quality on the field in realistic sporting scenarios. When athletes move optimally 'on the field' in realistic sport-specific conditions they can be considered to be 'movement ready' to RTS.

4 Fatigue

Although there is still no direct evidence that fatigue causes ACL injury, there is beginning to emerge good indirect evidence and a sound theoretical justification for its role in ACL injury, together with an association with numerous other injury types and athletic performance. Avoiding and being able to tolerate fatigue appear to be important considerations and should be recognised as vital components of ACL RTS training. Fatigue can manifest in many different ways and interact to affect injury risk and athletic performance [78].

Acute fatigue typically arises from acute periods of anaerobic activity during match play [86]. Although the research is equivocal [77, 78], experimentally simulated acute fatigue has resulted in reduced neuromuscular function and altered movement quality [79–83], some of which are considered 'at-risk ACL movements' [84, 85, 87, 88]. It also results in a decline in force-generating capacity of the muscles, preferentially affects RFD versus maximal strength [89] and can affect technical performance and decision-making ability

[90]. Thus, avoiding fatigue where possible is important for both athletic performance and general injury risk.

In terms of cumulative fatigue over the course of match play, it is apparent that a greater number of lower extremity injuries occur during the latter half or end of competitive match play [91, 92]. Over the course of match play there is a gradual decline in the ability to produce force, both maximally and explosively. Studies have reported deficits in maximal and explosive strength and power of the muscles of around 20% at the end of sporting match play [93–97]. Fatigue as a consequence of sporting match play appears to result in altered neuromuscular function and control, which may preferentially endanger the ACL (e.g. reduced knee flexor/extensor ratio) [98], although fatigue has yet to be definitively linked with ACL injury risk. Neuromuscular conditioning whilst fatigued has been shown to protect against declines in strength [99]. As such, exposing players to the required workloads of training and match play appears important to ensure they can tolerate these demands and limit excessive neuromuscular fatigue that could potentially endanger the ACL or other joints and/or muscles upon RTS.

Finally, cumulative fatigue over the course of multiple training sessions or matches could heighten athletes' risk of injury (any type of injury) [100]. Fatigue due to a heightened acute:chronic workload ratio (ACWR) that accompanies congested training and match play is a key risk factor for numerous injuries (see section 7), although not yet ACL injuries [101–105]. Managing players' loading and minimising exposure to sustained elevated loads appears important for injury prevention as a whole and likely important for ACL re-injury prevention. Across a range of sports, those with superior physical qualities are more resistant to injury [106–108]. In addition, stronger and fitter athletes are better able to tolerate elevated ACWR [108, 109], emphasising the importance of developing players' physicality to build resilience to fatigue. In addition, exposing players to fatigue during the rehabilitation period may also better prepare them to tolerate excessive training demands and limit strength loss associated with sporting activity upon RTS [99].

Important considerations in the management of fatigue appear to be (1) ensuring sufficient cardiovascular fitness to increase work capacity; (2) sufficient training prior to RTS to develop the ability to compete with the demands of match play and limit the cumulative fatigue over the course of match play (i.e. 90 minutes of work in competitive 11 v 11 football); and (3) sufficient exposure to sporting practice in a fatigued state so the athlete becomes accustomed to fatigue as well as training to avoid at-risk behaviours in this state. It appears that although an athlete may pass RTS criteria in a non-fatigued state (i.e. biomechanics, knee flexor/extensor ratio etc.), fatigue can result in altered neuromuscular function and control that may preferentially endanger the ACL (e.g. reduced hamstrings/quadriceps ratio, altered motor control) [96, 98]. As

such, testing in both a non-fatigued and fatigued state appears warranted when possible. This should ideally reflect the acute influence of fatigue from anaerobic work (e.g. the assessment of movement quality whilst acutely fatigued, such as following an intermittent running protocol) and assess the cumulative fatigue changes that result from training and match play (e.g. assessment of strength and movement quality at the end of or after a hard training session) where possible and plausible.

5 Sport-Specific Retraining

Sport-specific retraining represents an essential final phase of the ACL rehabilitation programme, prior to release back to sport. The RTS process should include a progression from on-field rehabilitation to return to team training and return to competition [109, 110], which is performed alongside sport-specific physical re-conditioning. On-field rehabilitation acts as a bridge from the rehabilitation and strength and conditioning environment to the return to sporting involvement with the team or coach. It aims to create a structured programme to guide an athlete back to training safely in the best physical condition to cope with the demands of sport. Importantly, it incorporates re-training of sport-specific movement, physical re-conditioning and restoring technical skills with a specifically developed programme. It is designed to simulate training in line with the health status of the athlete. This would be expected to have a positive impact on psychological readiness to RTS, which is becoming recognised as important for optimal and timely return to previous levels of performance [10, 111, 112]. Various factors such as poor conditioning, sport-specific movement patterns and the required sport-specific skills can affect performance and anxiety about RTS [112]. Separating on-field rehabilitation into distinct phases can provide a progressive increase in complexity (i.e. progression from controlled slow pre-planned movements through to highly chaotic reactive sport-specific movements) and volume and intensity of work. Successful progression from simple to more complex exercises should only occur when patients are fully confident and psychologically ready [113]. Those completing a greater proportion of their functional recovery programme on the field after serious knee injury reported superior RTS outcomes [114]. Furthermore, the completion of on-field rehabilitation was one of the specific criteria that resulted in lower re-injury risk upon RTS in professional footballers after ACLR [115]. However, it is important to note that no information about on-field rehabilitation was presented in this study [115] and the relation between completing on-field rehabilitation and re-injury risk in isolation was not documented. On-field rehabilitation typically involves simulated one versus one and small group training elements and does not fully prepare for the demands of most competitive

team sports. After completion of on-field rehabilitation, it is essential to include a period of full training with the team or coach, followed by gradual re-introduction to competitive match play, prior to full release to previous competitive level sport.

6 Has the Athlete Trained Enough to Return to Play Safely?

An often-neglected component of end-stage rehabilitation is ascertaining whether a player has played enough to cope with the demands of training and match play [103]. Load progression is a key part of rehabilitation and RTS decision making. It is becoming well established that inappropriate loading is a contributor to injury risk [101]. Specifically, athletes are at increased risk of injury when exposed to acute loads (typically 7 days of training load) in excess of their chronic training loads (typically 28 days of training load) [101–105, 116, 117]. In the context of late-stage rehabilitation and RTS, failure to develop sufficient chronic training loads prior to returning athletes back to sport would likely increase their injury risk [101], although there is limited actual evidence within the context of RTS.

7 Has the Athlete Restored Their Sport-Specific Fitness Profile?

A key part of RTS training is to prepare athletes for the specific demands of their sport. This is thought to facilitate optimal RTS, reduce re-injury risk and enhance performance upon RTS. However, often there is a lack of attention to sport-specific fitness training and ascertaining whether an athlete has sufficiently restored their sport-specific fitness profile.

Athletes need to have a sufficient physical profile to be able to contend with the demands of training and match play, as well as excel in their sports. Using football as an example, key physical skills are developed to a high level and may be decisive in football performance whereas other physical attributes need to meet a minimum requirement to be able to cope at a certain level [118–120]. The typical distance covered by top-level outfield male players during a game is 10–13 km [121–124]. During a game, players perform about 1200 discrete bouts of activity changing every 4–6 seconds [124], 150–250 brief, intense actions [119], with about 70 seconds between intense efforts [125], and 200–400 m of sprinting (distance covered over 7 m s^{-1}). They also perform numerous high intensity accelerations and decelerations (eight times as many accelerations as reported sprints per match), which although not resulting in speeds associated with high-intensity running are still metabolically taxing

[126, 127]. These explosive efforts involve challenging both the creatine phosphate and anaerobic glycolysis systems. Blood lactate concentrations recorded during football match play typically range from 2 to 12 mmol L⁻¹, with recorded individual values in excess of 12 mmol L⁻¹ [128]. The most decisive efforts leading to important outcomes/actions are anaerobic in nature and often involve a directional change [129]. As such, restoring explosive acceleration, deceleration and change of direction ability appears an essential element of RTS training.

During match play, players typically record average and peak heart rates of 85% and 98% maximum, respectively [130]. This represents an average of around 70% of maximal aerobic capacity (VO₂ max). Thus, players need to develop a very good aerobic and anaerobic cardiovascular capacity, specifically the ability to work for longer periods of time at high heart rates, to compete without the adverse effects of fatigue. Recent research indicates that football players fail to fully restore their aerobic fitness (measured as VO₂ max) 6 months following ACLR [131], indicating a greater need to prioritise and programme cardiovascular conditioning during the functional recovery period.

In summary, athletes need to be physically fit to cope with the demands of training and competition. Having a sport-specific reconditioning programme in place and ensuring an athlete has achieved their required physical profile (e.g. sport-specific physical testing) and completed sufficient work (e.g. performed the necessary work for training and or match play without adverse reactions) is important. The exact physical profile will depend on the player's sport, sex, position, standard of play and sporting profile (e.g. typically a fast player etc.). Physical performance testing can be performed as part of the RTS screening processes or alternatively when an athlete is safe to return to their team environment, as part of an analysis of training needs.

8 Implications for Late-Stage Rehabilitation and Return-to-Sport Training

The above sections highlight many aspects of late-stage rehabilitation and RTS training components that may be important for optimising the outcomes of athletes after ACLR. In order to optimise ACL outcomes and incorporate these elements, there is a need to update the current late-stage rehabilitation and RTS process. Recently, I have provided an update on the current RTS framework for athletes after sports injury [132]. In essence, the model describes the need to increase the focus of late-stage rehabilitation and RTS training towards performance re-training, with effective working relationships between the athlete's medical and performance teams. I also stressed the importance of having specific specialists in late-stage rehabilitation and

RTS training to effectively prepare athletes for RTS. The functional recovery process essentially involves a series of rehabilitation stages (e.g. early, mid- and late-stage rehabilitation), with a new continuum of RTS training consisting of on-field rehabilitation and gradual return to team training, competitive match play and performance [132].

This paper builds upon that model and describes the need to further strengthen the late-stage rehabilitation and RTS training process. To optimise ACL outcomes, there is a need to address the specific factors discussed within this paper. In order to achieve this, it is recommended to implement a specific five-staged late-stage rehabilitation and RTS training programme involving a combination of on-field rehabilitation, a movement re-training service and gym-based re-conditioning (Table 1). On-field rehabilitation acts as a bridge between the rehabilitation environment and team training. It should effectively prepare the athlete for the sport-specific movement, physical fitness, skill and loading demands (e.g. chronic training load) of their sport. In addition, on-field rehabilitation needs to be supported by an appropriate re-conditioning programme in the gym, in order to achieve the necessary function and physical conditioning to optimally RTS. This gym-based re-conditioning programme should focus on fully restoring maximal strength (isolated [e.g. knee extension], compound [e.g. leg press] and functional [e.g. squat]), explosive power performance (maximal power and RFD) and cardiovascular conditioning (e.g. an off-feet cardiovascular conditioning programme to complement the on-field conditioning programme), whilst targeted movement re-training using novel movement techniques, movement coaching and visual-motor re-training with bio-feedback techniques is needed to specifically address issues with movement quality. Collectively, these activities should address impairments associated with neuromuscular performance, movement quality and sport-specific performance to adequately prepare athletes for re-integration back into training. They should progress in volume, load, specificity and intensity towards that of team training, bridging the gap between rehabilitation and performance re-training.

In terms of optimal load progressions and management, there is excellent theoretical justification for inclusion of this concept in the RTS training and testing processes [101], but virtually no published research or even anecdotal experiences of how to do this in practice. Despite excellent theoretical merit for the ACWR in RTS training and testing [101], there is no published guidance for practitioners on how to optimally progress an athlete's loading during the RTS training process. Monitoring the external and internal training load of an athlete with global positioning system (GPS) technology for team sport players such as football players is recommended. The exact strategy in terms of load monitoring with GPS technology in the RTS process is not known. However, it would be advised

Table 1 Late-stage rehabilitation and on-field rehabilitation programme involving five stages after anterior cruciate ligament reconstruction for a football player with ambitions to return to play with the team

Late-stage rehabilitation and RTS periods					
	1	2	3	4	5
Entry criteria	No pain/swelling Symmetrical ROM Knee flexor and extensor LSI > 80% [114] Single leg-press strength > 1.25 × BM Ability to run at 8 km.h ⁻¹ for 10 minutes with sufficiently normalised running mechanics [114] Sufficient movement quality during foundation movements (e.g. squat [BL, OH, SL], lunge, hip hinge)	Same criteria as stage 1 Good unilateral landing control (forward and lateral) Good unilateral deceleration control (from forward and lateral running)	No pain/swelling Symmetrical ROM Knee flexor and extensor LSI > 90% Single leg-press strength > 2 × BM [35] Optimal movement quality 'pre-planned' sport-type tasks (movement analysis test > 85%) RFD LSI > 80%	Satisfactory progression through stage 3 on-field activity	Satisfactory progression through stage 4 on-field activity
Goal of period	Continue to restore muscle strength imbalances Core stability Functional strength Motor patterning Deceleration/landing capabilities	Multi-directional coordination Maximise strength Explosive strength and power	Reactive movement Sports technical gestures Physical conditioning	Sport-specific movement, skill training Conditioning in sport-specific drills	Training simulation/ re-conditioning
On-field activity	Uni-directional coordination Linear-based running (forward and lateral) Foundation movement tasks (e.g. lunging) Deceleration tasks in pre-planned situations of differing velocities Mobility drills	Multidirectional 'pre-planned' coordination drills (e.g. cutting and banded running drills, accelerations/decelerations) High speed linear-based running Easy technical drills CV (aerobic) conditioning using interval-based running	Speed and change of direction training Reactive movement re-training The sport-specific technical programme CV (aerobic/anaerobic) conditioning using interval-based running	Speed and agility training Sport-specific movement re-training Sport-specific 'skills' programme CV conditioning during sport-specific drills and multidirectional running tasks	Speed and agility training Sport-specific drills CV exposure under sport-specific training situations
Gym training	High load machine-based strengthening (5 RM) (e.g. leg press, knee extension) Moderate load 'functional' exercises (8–12 RM, squat, lunge, hip hinge) Bilateral plyometrics Core stability Off-feet cardiovascular fitness	High load machine-based strengthening (5 RM) Moderate load functional exercises (8–12 RM, squat, lunge, hip hinge) Core stability (load transfer) Off-feet anaerobic cardiovascular fitness UB, core endurance and aerobic fitness on recovery days	LB strength and power training Core strength/stability (load transfer) UB strength, core endurance and aerobic fitness on recovery days	LB strength and power training Core strength/stability (load transfer) UB strength, core endurance and aerobic fitness on recovery days	LB strength and power training Core strength/stability (load transfer) UB strength, core endurance and aerobic fitness on recovery days

Table 1 (continued)

Late-stage rehabilitation and RTS periods					
	1	2	3	4	5
Movement room (movement training with biofeedback)					
	Movement pattern re-training (e.g. squat etc.) Unilateral jumping and landing techniques Deceleration drills Bilateral plyometric techniques	Bilateral/unilateral power drills Unilateral plyometrics with biofeedback Change of direction drills			

There are five periods of progressively increasing intensity, volume, complexity and sport specificity. The programme is designed to prepare a football player for the demands of team training
BL bilateral, *BM* body mass, *CV* cardiovascular, *LB* lower body, *LSI* limb symmetry index, *OH* overhead, *RFD* rate of force development, *ROM* range of motion, *RTS* return to sport, *SL* single limb, *UB* upper body

to utilise the best available evidence within the literature. Based on Gabbett's research [104], it is recommended to assess the specific acute loading (7-day loading) on-field with GPS technology and contrast this with the athlete's chronic loading (28-day training load), ensuring that the ACWR stays below 1.5 during the RTS (this does not apply to the first few weeks of on-field activity when it is problematic ascertaining a chronic workload, as this takes 4 weeks, and the chronic training is essentially zero). Table 2 provides my current processes of progressive loading and management during the on-field reconditioning for professional football players aiming to return to full training with their team after ACLR. The metrics captured are based on the ability to understand if a player can cope with the different types of activity of their sport (see section 7). As football is an intermittent sport, the use of total distance and distance across various speed zones is needed alongside the ability to perform accelerations, decelerations and peak running speeds. In line with the need to develop sport-specific endurance and achieve the necessary intensity of activity to mimic training demands, heart rate (HR) measurement is important, with the minimum assessment being the time in aerobic (I use 70–85% HR maximum) and anaerobic (> 85% HR maximum) zones. A key aim to gradually progress the intensity and volume of activity towards that of at least 90% team training (or 45% of match activity), and achieve a 70% chronic training load (expressed as percentage of the player's normal training week, excluding match play) prior to return to team training (Table 2). Where GPS technology is not available, standardising the on-field rehabilitation programme (e.g. distinct periods of work and optimal progressions of drills) and recording the expected workload (e.g. drill 1, expected distance, high speed running exposure), whilst measuring the internal loading response via athlete session rating of perceived exertion and HR monitoring, is recommended.

In addition to load management progressions on the field, it is also essential to have load management strategies during the week to ensure optimal recovery and adaptation. This should involve variation of intensity from session to session with sufficient recovery between sessions, as well as balancing the training load between the field, gym and movement environment. I adopt an approach of varying on-field sessions between high quality and quantity sessions and hard, moderate and easy days. Quality sessions focus on optimal performance of the tasks at hand, with fewer repetitions and longer rest between efforts (to minimise possible fatigue). There is also a greater focus on sport-specific technical and tactical training during these sessions and they are designed to introduce new work or improve the quality of the work (e.g. movement or sports 'technique'). Quantity sessions involve greater workload, with more repetitions, less rest between repetitions and at higher intensities. They also aim

Table 2 An example of progressive loading and management during the five stages of the on-field rehabilitation period of an elite footballer prior to returning back to the team for training

On-field period	Peak speed (m s ⁻¹) (% maximum speed)	Total dist. (m)	HSR distance (m) (5.5–7 m s ⁻¹)	Sprint distance (m) (> 7 m s ⁻¹)	Acc distance (m) (> 3 m s ⁻²)	Dec distance (m > –3 m s ⁻²)	HR 70–85% max (min)	HR > 85% max (min)
1	4.5–6 (50–65)	3000–4500	0–100	0	0–15	0–40	0–10	0
2	6–7 (65–75)	4000–5000	100+	50	30+	50+	10–20	0–5
3	8+ (> 85)	4000–5000	200–400	100	50–100	50–100	25+	10–15
4	8.5+ (> 90)	4500+	300–500	100–150	100+	100+	20+	15+
5	8.5+ (> 90)	4500+	400–800	100–300	150+	150+	20+	20+

We adopt nine metrics with progression through stages based on achieving the desired intensity and quality of work, alongside other rehabilitation criteria such as movement quality, psychological readiness, strength and power, no pain or swelling etc

Acc acceleration, *Dec* deceleration, *HR* heart rate, *HSR* high-speed running, *m* meters, *max* maximum

Table 3 A typical training week for a professional football player in the final stages of the return-to-sport training programme prior to return to play with the team

Time	Mon	Tues	Wed	Thur	Fri	Sat	Sun
AM	<i>OFR—Quality</i> Technical/tactical Movement training Speed exposure	<i>OFR—Quantity</i> Conditioning Metric exposure Volume of activity	<i>Gym—Recovery</i> Massage Off-feet aerobic conditioning Upper-body strength Core endurance Flexibility	<i>OFR—Quality</i> Technical/tactical Movement training Speed exposure	<i>OFR—Quantity</i> Conditioning Metric exposure Volume of activity	<i>Gym—Recovery</i> Massage Off-feet aerobic conditioning Upper-body strength Core endurance Flexibility	<i>Off</i>
PM	<i>Movement room/gym</i> Neuromuscular control/motor patterning Lower-body power	<i>Gym</i> Lower body strength Core (load transfer/‘functional’)	<i>Off</i>	<i>Movement room/gym</i> Neuromuscular control/motor patterning Lower-body power	<i>Gym</i> Lower-body strength Core (load transfer/‘functional’)	<i>Off</i>	<i>Off</i>

The programme balances on-field re-training with different session types during the week (quality and quantity) with gym-based re-conditioning and recovery sessions

Words in italic indicate the environment and/or goal of the session

OFR on-field rehabilitation

to develop tolerance of higher training loads and stress the body system. This on-field rehabilitation programme is completed alongside a re-conditioning programme balancing the specific lower limb loading to provide a holistic approach to the functional needs of the athlete. This programme should be supported with recovery days using effective strategies (rest, massage, sleep, nutrition, upper-body strength training) [133, 134] the day after the ‘quantity’ sessions to enable effective recovery and support adaptation (see Table 3).

In summary, it is essential to have an optimal programme design in place, involving a progressive performance re-training model using on-field retraining and gym-based re-conditioning to gradually transition and re-enter an athlete back to team training. This programme should be followed by a period of team-based training and gradual re-introduction into competitive match play, with a long-term view of returning an athlete back to performance.

9 Implications for Return-to-Sport Testing

Ascertaining when an athlete is ready to RTS is complex. Although one cannot perfectly guarantee that an injury will not recur when the athlete returns back to sport, there are several factors that, if taken into consideration when determining return to competition, can help determine whether the risk of playing is acceptable. Recently, meeting RTS criteria prior to RTS has been shown to reduce the risk of re-injury by 75–84% [7, 115]. Having comprehensive rehabilitation and RTS processes in place, with criterion-based progressions and key discharge criteria, is thought to reduce the risk of re-injury and enhance outcomes after ACLR [7, 115]. However, it must be noted that passing RTS criteria by no means guarantees a player is safe upon RTS. In particular, in a recent prospective study of 158 professional male football players, those players failing to achieve the proposed RTS criteria were four times more likely to sustain a second

Table 4 Recommended testing to be included at the end of a football-specific on-field rehabilitation programme to support the decision-making process on player readiness to return to team training

Additional considerations for RTS	Suggested tests and criteria
Neuromuscular performance	Complete recovery of isolated muscle strength LSI 100% for quadriceps and hamstring strength (isokinetic or isometric test) knee flexor: extensor ratio > 0.60 [25, 114] Isometric RFD assessment of knee flexors and extensors (LSI > 90%) Leg press/squat assessment of isometric RFD (LSI > 90%) and peak force (LSI > 95%) Jump performance (CMJ) [143], drop jump and/or hopping performance] (LSI > 95%) [144]
Movement quality	High load qualitative movement evaluation (assessment in frontal and sagittal planes) of basic sports movements, ensuring correct mechanics (e.g. drop jump, hop tests, cut manoeuvre [planned and unplanned]). Other specific available movement quality tests include the LESS [145] Qualitative movement evaluation during sport-specific movement tasks on-field
Physical conditioning	Complete recovery of aerobic and anaerobic conditioning ($AT_2 > 11 \text{ km h}^{-1}$ and $AT_4 > 13.5 \text{ km h}^{-1}$ for professional footballers) [114] and/or Yo-Yo intermittent recovery test on-field [146] Complete recovery of speed (30-m acceleration test, peak speed using GPS data) and change of direction ability (505 agility or T-test)
Technical and tactical re-training	Complete a step-wise on-field rehabilitation programme, with increasing sport-specificity and complexity of technical and tactical re-training [114, 115]
Training load	Progressive loading with GPS monitoring and ACWR < 1.5 for each variable during the final week's on-field training at 90% of the normal acute week (7 days) with the team [104] Achieve the necessary chronic loading demands to cope with team training and subsequent match play (e.g. > 70% chronic training load across all relevant GPS and HR metrics) [101]

ACWR acute chronic workload ratio, *AT* anaerobic threshold, *CMJ* countermovement jump, *GPS* global positioning system, *HR* heart rate, *LESS* landing error score system, *LSI* limb symmetry index, *RFD* rate of force development, *RTS* return to sport

ACLR compared with those who met all proposed criteria (LSI > 90% for quadricep and hamstring muscle strength tests and three hop tests; t test < 11 s; and the completion of on-field sport-specific rehabilitation [115]). However, 12 of the 26 players with a second ACLR actually met the RTS criteria, while 28 of the 132 players with no second ACL injury did not pass RTS criteria. In a recent systematic review [135] of four published studies [7, 115, 136, 137], it was shown that passing RTS criteria does not significantly reduce the risk of second ACL injury, highlighting the limited ability of current RTS testing approaches to determine those who are at elevated risk of secondary ACL injury.

There is no consensus on when an athlete is ready to RTS, or the optimal testing procedure to determine sport readiness. Although having RTS criteria in place is thought important and optimises the RTS decision-making process, current RTS criteria appear to not be specific or sensitive enough to determine when a patient is really ready to RTS, particularly at an acceptable injury risk and performance level. Although suggesting an optimal test procedure goes beyond the scope of this review paper, the various factors discussed herein have relevance for RTS outcomes and subsequent re-injury risk. As such, their assessment as part of holistic RTS criteria is recommended. It is recommended that testing also consider other important and more commonly known factors that have been excluded from this review. These include the consideration of time [7, 13], psychological readiness [111, 112], knee status, confidence, stability and the more standard hop testing [138, 139], together

with medical approval and clearance by the supervising sports medicine physician and surgeon. On the basis of the findings from this review, it is proposed that researchers and practitioners also consider additional factors including understanding if the athlete has fully restored neuromuscular performance, movement quality, sport-specific fitness profile and specific loading requirements to cope with the demands of their sport (see Table 4 for an overview of recommended RTS testing for a professional football player).

It is important to also set acceptable risk tolerance [140] as part of the RTS decision-making process, and that this be dependent upon the specific contextual factors (injury, individual, sport etc.). However, given the devastating effects of secondary ACL re-injury, these risk tolerance levels should be set as low as realistically possible, whilst balancing the possibility of a risk averse approach and reduced patient satisfaction. Use of non-linear, multi-variate and complex systems thinking models, given the web of determinants that give rise to an injury, may be important and have a place in elite sport RTS diagnostics in the future [12, 140–142].

10 Conclusions

Outcomes following ACLR are not good enough. Successful recovery is likely influenced by multiple factors and some of the commonly overlooked or underrepresented factors have been discussed in this paper. Optimising neuromuscular function deficits (maximal isolated and global strength and

explosive neuromuscular performance), improving movement quality, ensuring sport-specific conditioning, minimising the possible detrimental effects of fatigue and ensuring an optimal RTS model could support better RTS success. Traditional functional recovery models and RTS testing typically fail to adequately restore or test these variables and it appears that more complete holistic interdisciplinary programmes and RTS testing are needed to enhance patient RTS outcomes after ACLR.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of interest Matthew Buckthorpe declares he has no conflicts of interest relevant to the content of this review.

References

- Marx RG, Jones EC, Angel M, Wickiewicz TL, Warren RF. Beliefs and attitudes of members of the American Academy of Orthopaedic Surgeons regarding the treatment of anterior cruciate ligament injury. *Arthroscopy*. 2003;19(7):762–70.
- Ardern CL, Webster KE, Taylor NF, Feller JA. Return to pre-injury level of competitive sport after anterior cruciate ligament reconstruction surgery: two-thirds of patients have not returned by 12 months after surgery. *Am J Sports Med*. 2011;39:538–43.
- Ardern CL. Anterior cruciate ligament reconstruction- not exactly a one-way ticket back to preinjury level: a review of contextual factors affecting return to sport after surgery. *Sports Health*. 2015;7(3):224–30.
- Lai CC, Ardern CL, Feller JA, Webster KE. Eighty-three per cent of elite athletes return to preinjury sport after anterior cruciate ligament reconstruction: a systematic review with meta-analysis of return to sport rates, graft rupture rates and performance outcomes. *Br J Sports Med*. 2018;52(2):128–38.
- Zaffagnini S, Grassi A, Marcheggiani Muccioli GM, et al. Return to sport after anterior cruciate ligament reconstruction in professional soccer players. *Knee*. 2014;21(3):731–5.
- Wiggins AJ, Granhi RK, Schneider DK, Stanfield D, Webster KE, Myer GD. Risk of secondary injury in younger athletes after anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Am J Sports Med*. 2016;44(7):1861–76.
- Grindem H, Snyder-Mackler L, Moksnes H, Engebretsen L, Risberg MA. Simple decision rules can reduce reinjury risk by 84% after ACL reconstruction: the Delaware-Oslo ACL cohort study. *Br J Sports Med*. 2016;50:804–8.
- Paterno MV, Rauh MJ, Schmitt LC, Ford KR, Hewett TE. Incidence of second ACL injuries 2 years after primary ACL reconstruction and return to sport. *Am J Sports Med*. 2014;42(7):1567–73.
- Webster KE, Feller JA. Exploring the high reinjury rate in younger patients undergoing anterior cruciate ligament reconstruction. *Am J Sports Med*. 2016;44(11):2827–32.
- Ardern CL, Kvist J, Webster KE. Psychological aspects of anterior cruciate ligament injuries. *Oper Tech Sports Med*. 2016;24(1):77–83.
- Ardern CL, Glasgow P, Schneiders A, et al. 2016 Consensus statement on return to sport from the first World congress in sports physical therapy, Bern. *Br J Sports Med*. 2016;50:853–64.
- Dingenen B, Gokeler A. Optimization of the return-to-sport paradigm after anterior cruciate ligament reconstruction: a critical step back to move forward. *Sports Med*. 2017;47(8):1487–500.
- Nagelli CV, Hewett TE. Should return to sport be delayed until 2 years after anterior cruciate ligament reconstruction? Biological and functional considerations. *Sports Med*. 2017;47(2):221–32.
- Buckthorpe M, Roi GS. The time has come to incorporate a greater focus on rate of force development training in the sports injury rehabilitation process. *Muscle Tendon Ligament J*. 2017;7(3):435–41.
- Krosshaug T, Nakamae A, Boden BP, et al. Mechanisms of anterior cruciate ligament injury in basketball: video analysis of 39 cases. *Am J Sports Med*. 2007;35:359–67.
- Beneke R, Taylor MJ. What gives Bolt the edge-A.V. Hill knew it already! *J Biomech*. 2010;43(11):2241–3.
- Thorstensson A, Karlsson J, Viitasalo HT, Luhtanen P, Komi PV. Effect of strength training on EMG of human skeletal muscle. *Acta Physiol Scand*. 1976;98:232–6.
- Angelozzi M, Madama M, Corsica C, et al. Rate of force development as an adjunctive outcome measure for return-to-sport decisions after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2012;42(9):772–80.
- Folland JP, Buckthorpe M, Hannah R. Human capacity for explosive force production: Neural and contractile determinants. *Scand J Med Sci Sports*. 2014;24(6):894–906.
- Maffiuletti NA, Aagaard P, Blazevich AJ, Folland J, Tillin N, Duchateau J. Rate of force development: physiological and methodological considerations. *Eur J Appl Physiol*. 2016;116(6):1091–116.
- Andersen LL, Aagaard P. Influence of maximal muscle strength and intrinsic muscle contractile properties on contractile rate of force development. *Eur J Appl Physiol*. 2006;96:46–52.
- Andersen LL, Andersen JL, Zebis MK, Aagaard P. Early and late rate of force development: differential adaptive responses to resistance training? *Scand J Med Sci Sports*. 2010;20(1):e162–9.
- Tillin NA, Pain MTG, Folland JP. Short-term unilateral resistance training affects the agonist-antagonist but not the force-agonist activation relationship. *Muscle Nerve*. 2011;43:375–84.
- Del Balso C, Cafarelli E. Adaptations in the activation of human skeletal muscle induced by short-term isometric resistance training. *J Appl Physiol*. 2007;103:402–11.
- Thomee R, Kaplan Y, Kvist Y, et al. Muscle strength and hop performance criteria prior to return to sports after ACL reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2011;19:1798–805.
- Wellsandt E, Fialia MS, Synder-Mackler L. Limb symmetry indexes can overestimate knee function after anterior cruciate ligament injury. *J Orthop Sports Ther*. 2017;47(5):334–8.
- Adams D, Logerstedt DS, Hunter-Giordano A, Axe MJ, Synder-Mackler L. Current concepts for anterior cruciate ligament reconstruction: a criterion-based rehabilitation progression. *J Orthop Sports Phys Ther*. 2012;42(7):601–14.
- Andersen JL, Aagaard P. Effects of strength training on muscle fiber types and size; consequences for athletes training for high-intensity sport. *Scand J Med Sci Sports*. 2010;20(2):32–8.
- Wallace BJ, Kernozek TW, White JM, et al. Quantification of vertical ground reaction forces of popular bilateral plyometric exercises. *J Strength Cond Res*. 2010;24(1):207–12.
- Aagaard P, Simonsen EB, Andersen JL, Magnusson P, Dyhre-Poulsen P. Increased rate of force development and neural drive of human skeletal muscle following resistance training. *J Appl Physiol*. 2002;93:1318–26.

31. Tillin NA, Pain MTG, Folland JP. Short-term training for explosive strength causes neural and mechanical adaptations. *Exp Physiol*. 2012;97:630–41.
32. Mangine GT, Hoffman JR, Wang R, et al. Resistance training intensity and volume affect changes in rate of force development in resistance-trained men. *Eur J Appl Physiol*. 2016;116:2367–74.
33. Van Cutsem M, Duchateau J, Hainaut K. Changes in single motor unit behaviour contribute to the increase in contraction speed after dynamic training in humans. *J Physiol*. 1998;513(Part 1):295–305.
34. Tillin NA, Folland JP. Maximal and explosive strength training elicit distinct neuromuscular adaptations, specific to the training stimulus. *Eur J Appl Physiol*. 2014;114(12):365–74.
35. Herrington L, Myer G, Horsley I. Task-based rehabilitation protocol for elite athletes following anterior cruciate ligament reconstruction: a clinical commentary. *Phys Ther Sport*. 2013;14(4):188–98.
36. Herrington L, Ghulam H, Comfort P. Quadriceps strength and functional performance after anterior cruciate ligament reconstruction in professional soccer players at time of return to sport. *J Strength Cond Res*. 2018. <https://doi.org/10.1519/jsc.00000000000002749> (Epub ahead of print).
37. Welling W, Benjaminse A, Seil R, et al. Low rates of patients meeting return to sport criteria 9 months after anterior cruciate ligament reconstruction: a prospective longitudinal study. *Knee Surg Sports Traumatol Arthrosc*. 2018. <https://doi.org/10.1007/s00167-018-4916-4> (Epub ahead of print).
38. Houglum PA. Therapeutic exercise for musculoskeletal injuries. Champaign: Human Kinetics; 2010.
39. Buckthorpe MW, Hannah R, Pain MTG, Folland JP. Reliability of neuromuscular measurements during explosive isometric contractions, with special reference to electromyography normalization techniques. *Muscle Nerve*. 2012;46(4):566–76.
40. Nuzzo JL, McBride JM, Cormier P, McCaulley GO. Relationship between countermovement jump performance and multijoint isometric and dynamic tests of strength. *J Strength Cond Res*. 2008;22(3):699–707.
41. Tillin NA, Pain MTG, Folland JP. Explosive force production during isometric squats correlates with athletic performance in rugby union players. *J Sports Sci*. 2013;31:66–76.
42. Hannah R, Folland JP, Smith SL, Minshull C. Explosive hamstrings-to-quadriceps force ratio of males versus females. *Eur J Appl Physiol*. 2015;115(4):837–47.
43. Struzik A, Juras G, Pietraszewski B, Rokita A. Effect of drop jump technique on the reactive speed index. *J Hum Kinet*. 2016;52:157–64.
44. Balsalobre-Fernandez C, Glaister M, Lockey RA. The validity and reliability of an iPhone app for measuring vertical jump performance. *J Sport Sci*. 2015;33(15):1574–9.
45. Decker MJ, Torry MR, Noonan TJ, Riviere A, Sterett WI. Landing adaptations after ACL reconstruction. *Med Sci Sports Exerc*. 2002;34(9):1408–13.
46. de Fontenay BP, Argaud S, Blache Y, Monteil K. Motion alterations after anterior cruciate ligament reconstruction: comparison of the injured and uninjured lower limbs during a single-legged jump. *J Athl Train*. 2014;49(3):311–6.
47. Goerger BM, Marshall SW, Beutler AI. Anterior cruciate ligament injury alters preinjury lower extremity biomechanics in the injured and uninjured leg: the JUMP-ACL study. *Br J Sports Med*. 2015;49:188–95.
48. Lee SP, Chow JW, Tillman MD. Persons with reconstructed ACL exhibit altered knee mechanics during high speed maneuvers. *J Sports Med*. 2014;35(6):528–33.
49. Paterno MV, Ford KR, Myer GD, Heyl R, Hewett TE. Limb asymmetries in landing and jumping 2 years following anterior cruciate ligament reconstruction. *Clin J Sports Med*. 2007;17(4):258–62.
50. Sterns KM, Pollard CD. Abnormal frontal plane knee mechanics during sidestep cutting in female soccer athletes after anterior cruciate ligament reconstruction and return to sport. *Am J Sports Med*. 2013;41(4):918–23.
51. Paterno MV, Schmitt LC, Ford KR, et al. Biomechanical measures during landing and postural stability predict second anterior cruciate ligament after anterior cruciate ligament reconstruction and return to sport. *Am J Sports Med*. 2010;38(10):1968–78.
52. Paterno MV, Kiefer AW, Bonnette S, et al. Prospectively identified deficits in sagittal plane hip-ankle coordination in female athletes who sustain a second anterior cruciate ligament injury after anterior cruciate ligament reconstruction and return to sport. *Clin Biomech*. 2015;30(10):1094–104.
53. Bien DP, Dubuque TJ. Considerations for late stage ACL rehabilitation and return to sport to limit re-injury risk and maximize athletic performance. *Int J Sports Phys Ther*. 2015;10(2):256–71.
54. Grooms DR, Page SJ, Nichols-Larsen DS, Chaudhari AM, White SE, Onate JA. Neuroplasticity associated with anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2017;47(3):180–9.
55. Swanik CB. Brains and sprains: the brain's role in non-contact anterior cruciate ligament injuries. *J Athl Train*. 2015;50(10):1100–2.
56. Padua DA, DiStefano LJ, Marshall SW, Beutler AI, de la Motte SJ, DiStefano MJ. Retention of movement pattern changes after a lower extremity injury prevention program is affected by program duration. *Am J Sports Med*. 2012;40(2):300–6.
57. Palmieri-Smith RM, Lepley LK. Quadriceps strength asymmetry following ACL reconstruction alters knee joint biomechanics and functional performance at time of return to activity. *Am J Sports Med*. 2015;43(7):1662–9.
58. Mills M, Frank B, Goto S, et al. Effect of restricted hip flexor muscle length on hip extensor muscle activity and lower extremity biomechanics in college-aged female soccer players. *Int J Sports Phys Ther*. 2015;10(7):946–54.
59. Zebis MK, Andersen LL, Bencke J, Kjaer M, Aagaard P. Identification of athletes at future risk of anterior cruciate ligament ruptures by neuromuscular screening. *Am J Sports Med*. 2009;37(10):1967–73.
60. Pollard CD, Sigward SM, Powers CM. ACL injury prevention training results in modification of hip and knee mechanics during a drop-landing task. *Orthop J Sports Med*. 2017;5(9):2325967117726267. <https://doi.org/10.1177/2325967117726267> (published online 2017 Sep 8).
61. Taylor JB, Waxman JP, Richter SJ, Shultz SJ. Evaluation of the effectiveness of anterior cruciate ligament injury prevention programme training components: a systematic review and meta-analysis. *Br J Sports Med*. 2015;49:79–87.
62. Bobbert MF, Van Soest AJ. Effects of muscle strengthening on vertical jump height: a simulation study. *Med Sci Sports Exerc*. 1994;26(8):1012–20.
63. Doyon J, Benali H. Reorganisation and plasticity in the adult brain during learning of motor skills. *Curr Opin Neurobiol*. 2005;15(2):161–7.
64. Gokeler A, Benjaminse A, Hewett TE, et al. Feedback techniques to target functional deficits following anterior cruciate ligament reconstruction: implications for motor control and reduction of second injury risk. *Sports Med*. 2013;43(11):1065–75.
65. Gokeler A, Benjaminse A, Welling W, et al. The effects of attentional focus on jump performance and knee joint kinematics in patients after ACL reconstruction. *Phys Ther Sport*. 2015;16:114–20.

66. Besier TF, Lloyd DG, Ackland TR, et al. Anticipatory effects on knee joint loading during running and cutting maneuvers. *Med Sci Sports Exerc.* 2001;33(7):1176–81.
67. Vanrenterghem J, Venables E, Pataky T, et al. The effect of running speed on knee mechanical loading in females during side cutting. *J Biomech.* 2012;45(14):2444–9.
68. Waldén M, Krosshaug T, Bjørneboe J, et al. Three distinct mechanisms predominate in non-contact anterior cruciate ligament injuries in male professional football players: a systematic video analysis of 39 cases. *Br J Sports Med.* 2015;49:1452–60.
69. Grooms D, Appelbaum G, Onate J. Neuroplasticity following anterior cruciate ligament injury: a framework for visual-motor training approaches in rehabilitation. *J Orthop Sports Phys Ther.* 2015;10(1):1–33.
70. Swanik CB, Covassin T, Stearne DJ, et al. The relationship between neurocognitive function and noncontact anterior cruciate ligament injuries. *Am J Sports Med.* 2007;35(6):943–8.
71. Dhillon MS, Kamal B, Sharad P. Differences among mechanoreceptors in healthy anterior cruciate ligaments and their clinical importance. *Muscles Ligaments Tendons J.* 2012;2(1):38–43.
72. Almonroeder TG, Garcia E, Kurt M. The effects of anticipation on the mechanics of the knee during single-leg cutting tasks: a systematic review. *Int J Sports Phys Ther.* 2015;10(7):918–28.
73. Brown SR, Brughelli M, Hume PA. Knee mechanics during planned and unplanned sidestepping: a systematic review and meta-analysis. *Sports Med.* 2014;44(11):1573–88.
74. Negahban H, Ahmadi P, Salehi R, et al. Attentional demands of postural control during single leg stance in patients with anterior cruciate ligament reconstruction. *Neurosci Lett.* 2013;556:118–23.
75. Negahban H, Hadian MR, Salavati M, et al. The effects of dual tasking on postural control in people with unilateral anterior cruciate ligament injury. *Gait Posture.* 2009;30(4):477–81.
76. Okuda K, Abe N, Katayama Y, et al. Effect of vision on postural sway in anterior cruciate ligament injured knees. *J Orthop Sci.* 2005;10(3):277–83.
77. Barber-Westin SD, Noyes FR. Effect of fatigue protocols on lower limb neuromuscular function and implications for anterior cruciate ligament injury prevention training: a systematic review. *Am J Sports Med.* 2017;45(14):3388–96.
78. Benjaminse A, Webster KE, Kimp A, Meijer M, Gokeler A. Revised approach to the role of fatigue in anterior cruciate ligament injury prevention: a systematic review with meta-analyses. *Sports Med.* 2019. <https://doi.org/10.1007/s40279-019-01052-6>.
79. De Ste Croix MB, Priestly AM, Lloyd RS, Oliver JL. ACL injury risk in elite female soccer: Changes in neuromuscular control of the knee following soccer specific fatigue. *Scand J Med Sci Sports.* 2015;25(5):e531–8.
80. Frank B, Gilsdorf CM, Goerger BM, Prentice WE, Padua DA. Neuromuscular fatigue alters postural control and sagittal plane hip biomechanics in active females with anterior cruciate ligament reconstruction. *Sports Health.* 2014;6(4):301–8.
81. Gokeler A, Eppinga P, Dijkstra PU, et al. Effect of fatigue on landing performance assessed with the landing error score system (less) in patients after acl reconstruction. A pilot study. *Int J Sports Phys Ther.* 2014;9(3):302–11.
82. Santamaria LJ, Webster KE. The effect of fatigue on lower-limb biomechanics during single-limb landings: a systematic review. *J Orthop Sports Phys Ther.* 2010;40:464–73.
83. Webster KE, Santamaria LJ, McClelland JA, Fellar JA. Effect of fatigue on landing biomechanics after anterior cruciate ligament reconstruction surgery. *Med Sci Sports Exerc.* 2012;44(5):910–6.
84. Chappell JD, Herman DC, Knight BS, Kirkendall DT, Garrett WE, Yu B. Effect of fatigue on knee kinetics and kinematics in stop-jump tasks. *Am J Sports Med.* 2005;33:1022–9.
85. Borotikar BS, Newcomer R, Koppes R, McLean SG. Combined effects of fatigue and decision making on female lower limb landing postures: central and peripheral contributions to ACL injury risk. *Clin Biomech.* 2008;23:81–92.
86. Mohr M, Krustup P, Bangsbo J. Match performance of high-standard soccer players with special reference to development of fatigue. *J Sports Sci.* 2003;21(7):519–28.
87. McLean SG, Fellin RE, Suedekum N, Calabrese G, Passerallo A, Joy S. Impact of fatigue on gender-based high-risk landing strategies. *Med Sci Sports Exerc.* 2007;39:502–14.
88. Sanna G, O'Connor KM. Fatigue-related changes in stance leg mechanics during sidestep cutting manoeuvres. *Clin Biomech.* 2008;23:946–54.
89. Buckthorpe M, Pain MT, Folland JP. Central fatigue contributes to the greater reductions in explosive than maximal strength with high-intensity fatigue. *Exp Physiol.* 2014;99(7):964–73.
90. Russell M, Benton D, Kingsley M. The effects of fatigue on soccer skills performed during a soccer match simulation. *Int J Sports Physiol Perform.* 2011;6(2):221–33.
91. Hawkins RD, Hulse MA, Wilkinson C, Hodson A, Gibson M. The association football medical research programme: an audit of injuries in professional football. *Br J Sports Med.* 2001;35:43–7.
92. Nagle K, Johnson B, Brou L, Landman T, Sochanska A, Comstock RD. Timing of lower extremity injuries in competition and practice in high school sports. *Sports Health.* 2017;9(3):238–46.
93. Greig M. The influence of soccer-specific fatigue on peak isokinetic torque production of the knee flexors and extensors. *Am J Sports Med.* 2008;36:1403–9.
94. Rahnama N, Reilly T, Lees A, Graham-Smith P. Muscle fatigue induced by exercise simulating the work rate of competitive soccer. *J Sports Sci.* 2003;21:933–42.
95. Rahnama N, Lees A, Reilly T. Electromyography of selected lower-limb muscles fatigued by exercise at the intensity of soccer match-play. *J Electromyogr Kinesiol.* 2006;16:257–63.
96. Van Melick N, van Rijn L, Nijhuis-van der Sanden MWG, Hoozeboom TJ, van Cingel REH. Fatigue affects quality of movement more in ACL-reconstructed soccer players than in healthy soccer players. *Knee Surg Traumatol Arthrosc.* 2018. <https://doi.org/10.1007/s00167-018-5149-2> (Epub ahead of print).
97. Small K, McNaughton L, Greig M, Lovell R. The effects of multidirectional soccer-specific fatigue on markers of hamstring injury risk. *J Sci Med Sport.* 2010;13(1):120–5.
98. Thorlund JB, Aagaard P, Madsen K. Rapid muscle force capacity changes after soccer match play. *Int J Sports Med.* 2009;30(4):273–8.
99. Small K, McNaughton L, Greig M, Lovell R. Effect of timing of eccentric hamstring strengthening exercises during soccer training: implications for muscle fatigability. *J Strength Cond Res.* 2009;23:1077–83.
100. Dupont G, Nedelec M, McCall A, et al. Effect of 2 soccer matches in a week on physical performance and injury rate. *Am J Sports Med.* 2010;38(9):1752–8.
101. Blanch P, Gabbett TJ. Has the athlete trained enough to return to play safely? The acute: chronic workload ratio permits clinicians to quantify a player's risk of subsequent injury. *Br J Sports Med.* 2016;50:471–5.
102. Bowen L, Gross AS, Gimpel M, Li FX. Accumulated workloads and the acute:chronic workload ratio relate to injury risk in elite youth football players. *Br J Sports Med.* 2017;51:452–9.
103. Duhig S, Shield AJ, Opar D, et al. Effect of high-speed running on hamstring strain injury risk. *Br J Sports Med.* 2016;50:1536–40.
104. Gabbett TJ. The training—injury prevention paradox: should athletes be training smarter and harder? *Br J Sports Med.* 2016;50:273–80.

105. Hulin BT, Gabbett T, Caputi P, Lawson DW, Sampson JA. Low chronic workload and the acute:chronic workload ratio are more predictive of injury than between-match recovery time: a two-season prospective cohort study in elite rugby league players. *Br J Sports Med.* 2016;50(16):1008–12.
106. Gabbett TJ, Ullah S, Finch CF. Identifying risk factors for contact injury in professional rugby league players—applicability of the frailty model for recurrent injury. *J Sci Med Sport.* 2012;15:496–504.
107. Malone S, Hughes B, Doran DA, Collins K, Gabbett TJ. Can the workload-injury relationship be moderated by improved strength, speed and repeated-sprint qualities? *J Sci Med Sport.* 2019;22(1):29–34.
108. Malone S, Roe M, Doran DA, Gabbett TJ, Collins K. Protection against spikes in workload with aerobic fitness and playing experience: the role of the acute:chronic workload ratio on injury risk in gaelic football. *Int J Sports Physiol Perform.* 2017;12:393–401.
109. Bizzini M, Hancock D, Impellizzeri F. Suggestions from the field for return to sports participation following anterior cruciate ligament reconstruction: soccer. *J Orthop Sports Phys Ther.* 2012;42(4):304–12.
110. English B. Phases of rehabilitation. *Foot Ankle Clin.* 2013;18(2):357–67.
111. Langford JL, Webster KE, Feller JA. A prospective longitudinal study to assess psychological changes following anterior cruciate ligament reconstruction surgery. *Br J Sports Med.* 2009;43:377–8.
112. Arden CL, Webster KE, Taylor NF, Feller JA. Return to sport following anterior cruciate ligament reconstruction surgery: a systematic review and meta-analysis of the state of play. *Br J Sports Med.* 2011;45(7):596–606.
113. te Wierike SC, van der Sluis A, van den Akker-Scheek I, Elferink-Gemser MT, Visscher C. Psychological factors influencing the recovery of athletes with anterior cruciate ligament injury: a systematic review. *Scand J Med Sci Sports.* 2013;23(5):527–40.
114. Della Villa S, Boldrini L, Ricci M, et al. Clinical outcomes and return-to-sports participation of 50 soccer players after anterior cruciate ligament reconstruction through a sport-specific rehabilitation protocol. *Sports Health.* 2012;4:17–24.
115. Kyritsis P, Bahr R, Landreau P, Miladi R, Witvrouw E. Likelihood of ACL graft rupture: not meeting six clinical discharge criteria before return to sport is associated with a four times greater risk of rupture. *Br J Sports Med.* 2016;50:946–51.
116. Soligard T, Schwelnlus M, Alonso J-M, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *Br J Sports Med.* 2016;50:1030–41.
117. Windt J, Zumbo BD, Sporer B, MacDonald K, Gabbett TJ. Why do workload spikes cause injuries, and which athletes are at higher risk? Mediators and moderators in workload–injury investigations. *Br J Sports Med.* 2017;51:993–4.
118. Bradley PS, Carling C, Gomez Diaz A, et al. Match performance and physical capacity of players in the top three competitive standards of English professional soccer. *Hum Mov Sci.* 2013;32(4):808–21.
119. Reilly T, Bangsbo J, Franks A. Anthropometric and physiological predispositions for elite soccer. *J Sports Sci.* 2000;18(9):669–83.
120. Tønnessen E, Hem E, Leirstein S, Haugen T, Seiler S. Maximal aerobic power characteristics of male professional soccer players, 1989–2012. *Int J Sports Physiol Perform.* 2013;8(3):323–9.
121. Mohr M, Krstrup P, Bangsbo J. Match performance of high-standard soccer players with special reference to development of fatigue. *J Sport Sci.* 2003;21:439–49.
122. Bangsbo J, Mohr M, Krstrup P. Physical and metabolic demands of training and match-play in the elite football player. *J Sports Sci.* 2006;24:665–74.
123. Mascio M, Bradley PS. Evaluation of the most intense high-intensity running period in English FA premier league soccer matches. *J Strength Cond Res.* 2013;27(4):909–15.
124. Stølen T, Chamari K, Castagna C, Wisloff U. Physiology of soccer: an update. *Sports Med.* 2005;35(6):501–36.
125. Bradley PS, Sheldon W, Wooster B, et al. High-intensity running in English FA Premier League soccer matches. *J Sports Sci.* 2009;27:159–68.
126. Osgnach C, Poser S, Bernardini R, Rinaldo R, di Prampero PE. Energy cost and metabolic power in elite soccer: a new match analysis approach. *Med Sci Sports Exerc.* 2010;42(1):170–8.
127. Varley MC, Aughey RJ. Acceleration profiles in elite Australian soccer. *Int J Sports Med.* 2013;34(1):34–9.
128. Krstrup P, Mohr M, Steensberg A, et al. Muscle and blood metabolites during a soccer game: implications for sprint performance. *Med Sci Sports Exerc.* 2006;38:1165–74.
129. Faude O, Koch T, Meyer T. Straight sprinting is the most frequent action in goal situations in professional soccer. *J Sports Sci.* 2012;30:625–31.
130. Bangsbo J. The physiology of soccer—with special reference to intense intermittent exercise. *Acta Physiol Scand.* 1994;151(suppl. 619):1–155.
131. Almeida AM, Santos Silva PR, Pedrinelli A, Hernandez AJ. Aerobic fitness in professional soccer players after anterior cruciate ligament reconstruction. *PLoS One.* 2018;13(3):e0194432. <https://doi.org/10.1371/journal.pone.0194432> (eCollection 2018).
132. Buckthorpe M, Frizziero A, Roi GS. Update on functional recovery process for the injured athlete: return to sport continuum redefined. *Br J Sports Med.* 2018. <https://doi.org/10.1136/bjsports-2018-099341> (published online first: 29 September 2018).
133. Dupuy O, Douzi W, Theurot D, Bosquet L, Duqué B. An evidence-based approach for choosing post-exercise recovery techniques to reduce markers of muscle damage, soreness, fatigue, and inflammation: a systematic review with meta-analysis. *Front Physiol.* 2018;9:403. <https://doi.org/10.3389/fphys.2018.00403> (eCollection 2018).
134. Reilly T, Ekblom B. The use of recovery methods post-exercise. *J Sports Sci.* 2005;23(6):619–27.
135. Losciale JM, Zdeb RM, Ledbetter L, Reiman MP, Sell TC. The association between passing return-to-sport criteria and second anterior cruciate ligament injury risk: a systematic review with meta-analysis. *J Orthop Sports Phys Ther.* 2019;49(2):43–54. <https://doi.org/10.2519/jospt.2019.8190> (Epub 2018 Nov 30).
136. Nawasreh Z, Logerstedt D, Cummer K, et al. Do patients failing return-to-activity criteria at 6 months after anterior cruciate ligament reconstruction continue demonstrating deficits at 2 years? *Am J Sports Med.* 2017;45:1037–48. <https://doi.org/10.1177/0363546516680619>.
137. Sousa PL, Krych AJ, Cates RA, et al. Return to sport: does excellent 6-month strength and function following ACL reconstruction predict midterm outcomes? *Knee Surg Sports Traumatol Arthrosc.* 2017;25:1356–63. <https://doi.org/10.1007/s00167-015-3697-2>.
138. Davis GJ, McCarthy E, Provencher M, Manske RC. ACL return to sport guidelines and criteria. *Curr Rev Musculoskeletal Med.* 2017;10:307–14.
139. Noyes FR, Barber SD, Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture. *Am J Sports Med.* 1991;19(5):513–8.
140. Shier I. Strategic Assessment of Risk and Risk Tolerance (StARRT) framework for return-to-play decision-making. *Br J Sports Med.* 2015;49:1311–5.

141. Bittencourt NFN, Meeuwisse WH, Mendonca LD, et al. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition-narrative review and new concept. *Br J Sports Med.* 2016;50:1308–14.
142. Gokeler A, Welling W, Zaffagnini S, Seil R, Padua D. Development of a test battery to enhance safe return to sports after anterior ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc.* 2017;25(1):192–9.
143. O'Malley E, Richter C, King E, et al. Countermovement jump and isokinetic dynamometry as measures of rehabilitation status after anterior cruciate ligament reconstruction. *J Athl Train.* 2018;53(7):687–95.
144. Noyes FR, Barber SD, Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture. *Am J Sports Med.* 1991;19:513–8.
145. Padua DA, Marshall SW, Boling MC, et al. The Landing Error Scoring System (LESS) is a valid and reliable clinical assessment tool of jump-landing biomechanics: The JUMP-ACL study. *Am J Sports Med.* 2009;37:1996–2002.
146. Bangsbo J, Iaia FM, Krstrup P. The Yo-Yo intermittent recovery test: a useful tool for evaluation of physical performance in intermittent sports. *Sports Med.* 2008;38(1):37–51.